

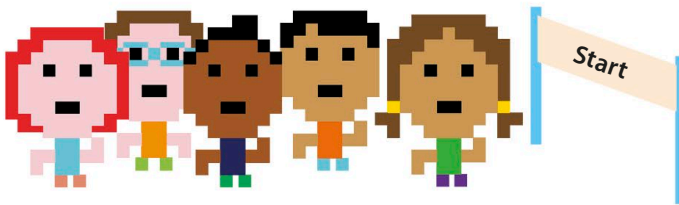


◁ Runaway turtle!

Given complete freedom to roam, the turtle tends to wander out of the window. As you put the program together, you'll write some code to check on the turtle's position and stop it from straying too far. Otherwise, this will turn into a vanishing turtle project!

Getting started

Start by setting up and saving a new file, importing the modules that the program will need, and making a couple of useful functions to get user input.



1 Create new files

Open IDLE and create a new file. Save it as "rainbow.py".

2 Add the modules

Type these two lines at the top of your file to import the `turtle` and `random` modules. Remember to use `import turtle as t`, so that you don't have to type the word "turtle" every time you want to use a function from the Turtle module. You can just call it `t`.

```
import random
import turtle as t
```

3 Assign line length

Next make a function that will let the user decide whether the turtle paints long, medium, or short lines. You won't use it until Step 4, but this will get the program ready for when you need it. Type this bit of code beneath the code in Step 1.

```
import turtle as t
```

```
def get_line_length():
```

```
    choice = input('Enter line length (long, medium, short): ')
```

```
    if choice == 'long':
```

```
        line_length = 250
```

```
    elif choice == 'medium':
```

```
        line_length = 200
```

```
    else:
```

```
        line_length = 100
```

```
    return line_length
```

This asks the user to choose how long the line is.

This command passes `line_length` back to the code that called this function.

For a short line, set `line_length` to 100.